

Technical Document #955: Database Systems - Comprehensive Overview

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Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Database connection pooling reuses database connections to reduce overhead. It improves performance for applications with many concurrent users.

Database indexing improves query performance by creating data structures that allow faster data retrieval.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Key Takeaways:

- Database monitoring tracks performance metrics and identifies bottlenecks.
- Database replication copies data across multiple servers.
- Database migration tools manage schema changes and data transformations.